Capstone Assignment: Supervised-Learning-Readiness EDA

GA 5800 — Exploratory Data Analysis Capstone | Summer 2026

Due: May 21, 2026 at 11:59 pm **Weight:** 10% of final grade **Format:** Executed Jupyter notebook + Data Card + Readiness Memo **Submit to:** [Gradescope / Canvas link]

If your Stakeholder project is a **classification** problem follow the asks below for classification; if your project is a **regression** problem follow the asks for regression.

Important Note: You may encounter EDA techniques, statistical concepts, or Python libraries that weren't covered in your classes. That's intentional. Part of this assignment is figuring out how to use unfamiliar tools — look things up, read the docs, and make them work.

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1. Scenario

You have just been hired as the lead analyst on a modeling project. The modeling team will build a supervised model on our Stakeholder's dataset, but before they touch it they need a thorough, evidence-based, end-to-end picture of the data: what it contains, what it is missing, what signals it carries with respect to the target, what risks it poses (leakage, drift, multicollinearity, fairness), and exactly how each feature should be handled in the modeling pipeline.

This exercise is that picture.

You are not building a model. You are producing the **artifact a senior data scientist would demand before allowing a model to be trained.** Your work will be judged on whether a competent modeler could pick up your notebook, data card, and memo and immediately make defensible preprocessing and modeling decisions.

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2. Dataset

Item	Value
Dataset	`[DATASET NAME]`
Source / provenance	`[SOURCE]`
Approximate size	`[ROWS × COLS]`
Target variable	`[TARGET]`
Supervised task	`[CLASSIFICATION / REGRESSION]`
Prediction unit	`[ROW = ?]`
Provided files	`[FILE LIST]`

(If any field above is "TBD", confirm in the first cell of your notebook.)

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3. Learning Objectives

By submission, you will have demonstrated the ability to:

1. Frame a raw dataset as a supervised-learning problem with an explicit target, prediction unit, and evaluation metric.
2. Audit data quality and missingness with both descriptive and inferential evidence.
3. Select **statistically appropriate** univariate, bivariate, and multivariate analyses based on each variable's measurement type.
4. Quantify feature–target signal using both linear (correlation) and non-linear (mutual information) tools, and reconcile their disagreements.
5. Detect and document leakage, multicollinearity, and outliers as actionable modeling risks.
6. Translate exploratory findings into a concrete, per-feature preprocessing plan and a defensible train/validation/test split strategy.
7. Communicate readiness to a modeling audience in writing.

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4. The 10-Stage Framework

You must complete all ten stages **in order**, in a single notebook, with clearly numbered section headers (`## Stage 1: …`). Each stage requires both *analysis* and *written interpretation* — code without interpretation will not earn full credit.

> **Statistical-test selection rule (applies throughout):** the choice of statistic must match the measurement scale of the variables involved. Indiscriminately applying Pearson correlation to ordinal or skewed variables, or `mutual_info_regression` to a classification target, will be treated as a methodological error.

Stage 1 — Problem Framing & Target Definition *(5 pts)*

Produce:

- One-paragraph problem statement: what would a model predict, for whom, and why?
- Explicit target definition: variable name, type (binary / multiclass / continuous / count / ordinal), unit of observation, time-of-prediction (what is known *at prediction time* vs. observed afterward — this matters for leakage).
- A proposed primary evaluation metric **with justification** (e.g., why F1 vs. ROC-AUC vs. PR-AUC; why RMSE vs. MAE vs. MAPE).
- A short statement of whether this is a single-target, multi-target, or multi-output problem.

Stage 2 — Data Inventory & Schema Audit *(7 pts)*

Produce:

- Shape, memory footprint, dtype table.

- A **semantic feature-type table** (one row per column) with these columns: `name | pandas dtype | semantic type | role | notes` where `semantic type ∈ {continuous, count, ordinal, nominal, binary, datetime, text, identifier, geographic, other}` and `role ∈ {feature, target, identifier, metadata, leakage-suspect}`.
- A list of columns where the pandas dtype disagrees with the semantic type (e.g., zip codes stored as `int64`, dates stored as `object`) and your planned coercion.

Required output: a markdown or DataFrame table; this table is the seed of your final Data Card.

Stage 3 — Data Quality Assessment *(8 pts)*

Produce:

- Exact and approximate duplicate-row counts (justify your "approximate" key).
- Constant and quasi-constant columns (variance ≈ 0 or single value > 95%).
- ID-like columns (cardinality ≈ row count).
- Impossible / out-of-domain values (negatives where impossible, dates in the future, percentages > 100, etc.). For each, propose: drop / cap / impute / flag.
- Encoding glitches (mojibake, mixed case in categories, leading/trailing whitespace, inconsistent units).
- For each issue: severity (low / medium / high) and proposed remediation.

Stage 4 — Missingness Analysis *(8 pts)*

Produce:

- Per-column missing count and percentage; total row-completeness distribution.
- Missingness matrix and missingness correlation heatmap (e.g., `missingno`).
- **Missingness vs. target test**: for each column with non-trivial missingness, test whether missingness is associated with the target (chi-square if classification target, Mann-Whitney / Kruskal-Wallis if regression target). Report effect size, not just p-value.
- A reasoned MCAR / MAR / MNAR posture for the most-missing columns. You do not need to *prove* the mechanism — you need to argue it.
- A per-column imputation plan (mean / median / mode / model-based / explicit "Missing" category / row-drop) **with justification.**

Stage 5 — Univariate Analysis *(8 pts)*

Numeric features:

- Summary statistics including count, mean, median, std, min, max, IQR, skewness, kurtosis for all features.
- Distribution plots (histogram + KDE, or ECDF) for at least 8 numeric features.

Categorical features:

- Cardinality table.
- Frequency tables for low-cardinality categoricals; **rare-level audit** for high-cardinality categoricals (define your "rare" threshold and propose grouping into "Other").

- Identify any categoricals masquerading as numeric (e.g., zip codes).

Stage 6 — Target Analysis *(10 pts)*

Classification target:

- Class frequencies and proportions; bar chart.
- Imbalance ratio; recommended response (none / class weights / resampling / threshold tuning) with justification tied to the metric chosen in Stage 1.
- Stratification implications for your split strategy (Stage 10).

Regression target:

- Distribution, summary statistics, skew, kurtosis.
- Outlier and zero-inflation check.
- Transformation evaluation (e.g., log-target) with before/after plots; recommend whether to model on transformed scale.

Both:

- If a temporal axis exists, plot target over time and comment on drift or seasonality.

Stage 7 — Bivariate Signals vs. Target *(12 pts)*

For **every** feature retained after Stage 3, quantify its bivariate relationship with the target using a test appropriate to the variable-type pair:

Feature type	Classification target	Regression target
Continuous / count	Point-biserial r; ANOVA F or Kruskal-Wallis H; AUC of feature alone	Pearson r, Spearman ρ; report both
Ordinal	Mann-Whitney / Kruskal-Wallis; Spearman ρ	Spearman ρ, Kendall τ
Nominal	Chi-square + **Cramér's V**; target rate by level	One-way ANOVA + $\boldsymbol{\eta^2}$; group means with CIs
Binary	Two-proportion z; difference in target rate	t-test or Mann-Whitney; Cohen's d

Required output:

- A ranked table of all features with: test used, statistic, p-value, **effect size**, and one-line interpretation.
- At least 6 illustrative plots (e.g., violin/box-by-class, target-rate-by-bin, scatter with LOWESS).
- Explicit acknowledgement of multiple-comparison risk (Bonferroni, Benjamini-Hochberg, or a stated "we are screening, not confirming" caveat).

> A statistically significant p-value with a trivial effect size is **not a signal**. Effect size is required.

Stage 8 — Multivariate Structure *(12 pts)*

Produce:

- Pearson **and** Spearman correlation matrices for numeric features (heatmaps). Comment on disagreements (linear vs. monotonic).
- Multicollinearity assessment via **VIF** (or condition number) on the candidate numeric feature set; flag any VIF > 10.
- Pairplots or facetted scatter for the top 6–8 features identified in Stage 7, colored/grouped by target.
- Optional but encouraged: PCA scree plot and 2-component projection colored by target — interpret what (if anything) the principal axes mean.
- **Multivariate outlier detection**: Isolation Forest *or* Mahalanobis distance on the numeric feature set. Show the score distribution; investigate the top 1% — are these errors, edge cases, or a meaningful subpopulation?

Stage 9 — Mutual Information & Leakage Screen *(12 pts)*

Mutual information:

- Compute MI between every feature and the target using the appropriate scikit-learn estimator (`mutual_info_classif` for classification, `mutual_info_regression` for regression). Encode categoricals appropriately and pass the `discrete_features` mask correctly.
- Rank features by MI; produce a horizontal bar chart of the top 20.
- Compare the MI ranking with the linear-association ranking from Stage 7. **Identify and discuss at least three features where the two rankings disagree** — this is where non-linear signal lives.

Leakage screen (required, graded heavily): For every feature in the top quartile of MI *or* top quartile of bivariate effect size, ask and answer in writing:

1. Is this feature available **at the time of prediction**, or is it observed only after the target is realized?
2. Is it derived from, or a near-restatement of, the target?
3. Is it an identifier, timestamp, or post-event flag?
4. Does it look "too good to be true" given domain context?

Any feature that fails 1–4 must be flagged as **leakage-suspect** in your Data Card and excluded from the modeling set.

Sanity check (optional): Train a *single* tree-based "EDA model" (e.g., `RandomForestClassifier` / `RandomForestRegressor` with default-ish hyperparameters and a basic preprocessor) **purely as an importance probe**. Report permutation importance on a held-out split. This is **not** model selection — it is a non-linear sanity check on the MI ranking. State this caveat in writing.

Stage 10 — Supervised-Learning Readiness Deliverables *(15 pts)*

This is the payoff stage. Produce **all four** of the following:

10a. Data Card / Feature Dictionary (table). One row per feature, columns: `name | semantic type | % missing | univariate notes | bivariate signal (stat, effect size) | MI | multicollinearity flag | leakage flag | recommended treatment | encoding | scaling | keep/drop`

10b. Train / Validation / Test split recommendation. Explicitly choose and justify one of: random, stratified, grouped (specify grouping key), time-based (specify cutoff), or nested. Address: leakage between splits, target stratification, group leakage, temporal validity. Specify the fractions.

10c. Preprocessing pipeline sketch. A diagram or itemized list mapping feature → transformer → output, in the form a `sklearn.pipeline.Pipeline` / `ColumnTransformer` would consume. You do not need to write the pipeline code; you must specify it precisely enough that someone else could.

10d. Top-10 candidate predictors + Known Risks list. A short ranked list of the ten features most likely to carry signal, citing the evidence (Stage 7 effect size + Stage 9 MI + multivariate behavior). Plus a bulleted **Known Risks** list (leakage suspects retained or removed, multicollinearity clusters, missingness assumptions, drift concerns, fairness-sensitive features and proxies).

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5. Required Submission Artifacts

Submit **three files** as a single zipped folder named `lastname_firstname_capstone.zip`:

1. **`lastname_firstname_capstone_eda.ipynb`** — fully executed notebook, all 10 stages, all cells run, outputs visible. Set `np.random.seed(42)` (or document your seed) at the top.
2. **`lastname_firstname_data_card.md`** — the Stage 10a feature dictionary as a standalone markdown table (also embedded in the notebook).
3. **`lastname_firstname_readiness_memo.pdf`** — 3–5 pages, written for the modeling team. Sections: *(i)* problem framing, *(ii)* dataset health summary, *(iii)* top signals and why we believe in them, *(iv)* leakage and risk register, *(v)* split & preprocessing recommendation, *(vi)* open questions for the modeling team.

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6. Quality Standards

Element	Standard
Visualization	Every plot has title, axis labels with units, legend if needed, and a caption explaining the takeaway (not just a description). Color choices are colorblind-safe.
Statistical rigor	Tests match measurement scale. Effect sizes accompany every p-value. Multiple-comparison risk is acknowledged. Assumptions of each test are stated.
Writing	Findings are written as claims with evidence ("Cramér's V = 0.34 between `region` and `target` indicates a moderate association; target rate ranges from 12% (West) to 41% (Northeast)"), not as descriptions of code.
Reproducibility	Random seed set. Data path documented. Notebook runs top-to-bottom without manual intervention. Environment is `environment/requirements.txt`-compatible.
Code quality	No dead code. No printed `head()` calls left in for debugging. Long analyses factored into functions where clarity benefits.

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7. Embedded Grading Rubric (100 points)

7.1 Stage Points

Stage	Topic	Points
1	Problem framing & target definition	5
2	Data inventory & schema audit	7
3	Data quality assessment	8
4	Missingness analysis	8
5	Univariate analysis	8
6	Target analysis	10
7	Bivariate signals vs. target	12
8	Multivariate structure	12
9	Mutual information & leakage screen	12
10	SL-readiness deliverables	15
—	**Subtotal**	**97**
—	Reproducibility & code quality	1
—	Statistical rigor (cross-cutting)	1
—	Writing & visualization quality	1
—	**Total**	**100**

7.2 Anchor Descriptors (apply to every stage)

Level	Description
Exemplary (90–100% of stage pts)	All required elements present. Methods correctly matched to data types. Effect sizes accompany inferential claims. Interpretation goes beyond restating numbers — it explains *why* and *what to do about it*. Readiness implications are explicit.
Proficient (75–89%)	All required elements present with minor methodological imperfections (e.g., one mismatched test, missing one effect size, incomplete remediation plan). Interpretation is correct but shallow in places.
Developing (60–74%)	One or more required elements missing or substantively wrong. Tests applied without regard to scale, or p-values reported without effect sizes. Interpretation reads like narration of code output.
Insufficient (< 60%)	Stage is largely missing, copied generically without engaging with this dataset, or contains methodological errors that invalidate the conclusions.

7.3 Automatic Deductions (applied after stage scoring; can be stacked)

Issue	Deduction
Notebook does not execute top-to-bottom	−10

Issue	Deduction
No random seed set	−2
Plots without titles or axis labels (any)	−2 per offense, max −6
Bivariate stage relies only on Pearson without acknowledging type-appropriateness	−4
`mutual_info_regression` used on a classification target (or vice versa)	−5
Leakage-suspect feature retained without flagging in Data Card	−6
No train/val/test split recommendation in Stage 10	−5
Memo missing or < 2 pages	−5
Data Card missing	−5
Late: 1–2 days	−10%
Late: 3–7 days	−20%
Late: > 7 days	See instructor

7.4 Bonus (up to +5, cannot push above 100)

- **+2** for a thoughtful fairness / proxy-variable section.
- **+2** for credible drift / temporal-stability analysis when the dataset has a time axis.
- **+1** for an interactive exploratory artifact (e.g., Plotly dashboard cell) that adds insight, not decoration.

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8. Submission Checklist

- [ ] Notebook executes top-to-bottom on a fresh kernel.
- [ ] Random seed set and documented.
- [ ] All 10 stages have numbered headers.
- [ ] Data Card produced as both notebook table and standalone `.md`.
- [ ] Readiness Memo PDF, 3–5 pages, addresses the six required sections.
- [ ] Every p-value is accompanied by an effect size.
- [ ] Every plot has title, labels, and caption.
- [ ] Leakage screen explicitly addressed for every top-quartile feature.
- [ ] Train/val/test split strategy stated and justified.
- [ ] Preprocessing pipeline sketch present.
- [ ] Files named per convention; submitted as zipped folder.

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9. Tips for Success

1. **Start with Stages 1, 2, and 10's Data Card skeleton.** Treat the Data Card as a living document; fill it in as you complete each stage. By the time you reach Stage 10, two-thirds of it will already be written.
2. **Pick the right test, then run it.** Reach for documentation before reaching for `.corr()`.

3. **An effect size is a claim about magnitude.** A p-value is a claim about noise. You need both.
4. **Be paranoid about leakage.** If a feature looks magical, it usually is — and not in a good way.
5. **Write the memo last but plan it first.** Knowing what the modeling team needs to hear shapes what evidence you collect.
6. **Read your own notebook as a stranger would.** Anything that needs verbal explanation in office hours needs to be in the markdown.

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10. Late Submission Policy

When	Penalty
On time	Full credit
1–2 days late	−10%
3–7 days late	−20%
> 7 days late	See instructor

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11. Academic Integrity

You may use language models and online references to look up syntax, statistical definitions, and library APIs. You may **not** submit AI-generated narrative interpretation, AI-generated memo prose, or AI-generated leakage analysis as your own. The interpretive judgment in this assignment — *what does this mean, what should we do, what are we risking* — is the work being graded. Cite any reference you used.

Questions? CT Husky Discussion Board:
Submission link: [Gradescope / Canvas link]